The University of Lahore
Department of Computer Science & IT
Final Term Examination

Fall-2023

Course Title:	Computer Organization & Assembly Language	Course Code:	CS11207	Credit Hours:	**4**
Course Instructors:	Mr. M. Yasin, Ms. Asmara, Ms.Uzma, Mr.Mutahher, Ms.Zarbakht, Mr.Asif	Program Name:	**BSCS**		
Time Allowed:	**120 Minutes**	**Maximum Marks:**	**40**		
Date:	**04-01-24**	Course Mentor:	**Mr. M. Yasin Nasir**		
Student's Name:		Signature:			
Reg. No:		Section:	**A,B,C,D,E,F,G,I,J**		

Important Instructions / Guidelines:

- Do not forget to pray before starting to attempt the paper. Trust me it helps.
- **Remember! SOMEONE is always with you (Be Relaxed), and HE is also watching you (Be Honest)**
- Question Paper is **SELF EXPLANATORY**. Understanding the Question Paper is part of Solution Solve subjective part on the answer sheet provided separately.
- Your paper will be marked as **ZERO**, if involved in any kind of cheating.
- **Attempt all the questions on question paper except Q4, Q5 & Q6.**

Question # 1: Find the output of the following questions. (5 marks)

**.MODEL SMALL
.STACK 100H
.DATA
A DB 1, 2, 3, 4
B DB 6, 7, 8, 9
.CODE
MAIN PROC
MOV AX, @DATA
MOV DS, AX
MOV CX, 0000
MOV CL, 03
LEA BP, A
LEA SI, B
L1:
MOV DH, [BP]
MOV DL, [SI]
MOV [BP], DL
MOV [SI], DH
MOV DH, [BP]
MOV DL, [SI]**
	**MOV DL,DH
	ADD DL,'0'
	MOV AH,02
	INT 21h
	INC BP
	INC SI
	DEC CL
	CMP CL, 00
	JNZ L1
	MOV AH, 4CH
	INT 21H
	END MAIN**
		Output:

Question # 2: Dry run the following code for the given values as decimal input. (Note: INDEC & OUTDEC procedure returns decimal input in AX. You don't need to write INDEC and OUTDEC procedures) (6 marks)

```
.MODEL SMALL
.STACK 100H

 .DATA
   PROMPT_1  DB  'Enter a number : $'
   PROMPT_2  DB  0DH,0AH,'Resultant: $'

 .CODE
   MAIN PROC
     MOV AX, @DATA
     MOV DS, AX
     LEA DX, PROMPT_1
     MOV AH, 9
     INT 21H
     CALL INDEC
     CALL UNKNOWN
     PUSH AX
     LEA DX, PROMPT_2
     MOV AH, 9
     INT 21H
     POP AX
     CALL OUTDEC
     MOV AH, 4CH
     INT 21H
   MAIN ENDP

UNKNOWN PROC
   ; Input: AX

   MOV CX, AX
   MOV AX, 1
   @LOOP:
     MUL CX
   LOOP @LOOP
   RET
UNKNOWN ENDP

END MAIN
```

Dry-run for following values:

INDEC Values	Output of the program (OUTDEC)
3	
4	
5	

Space for rough work.

Question # 3: **(6 Marks)**

Assume that $IP = 100H,\ SS = 07EF$

- **Find the contents of the stack for STACK_OPERATIONS .**
- **Fill the data in given stacks with appropriate Logical Addressing, Stack data values and SP.**
- **Show the sequence of instructions needed at the empty lines in the program to restore all the registers to their original values.**

```
.MODEL SMALL
.STACK 0F2Eh
.DATA
.CODE
MAIN PROC
MOV AX, @DATA
MOV DS, AX
        MOV AX,3291H
        MOV BX,0F43CH
        MOV CX,09H

        CALL STACK_OPERATIONS

MOV AH,4CH
INT 21H
MAIN ENDP
END MAIN
```

```
STACK_OPERATIONS PROC

PUSH AX
MOV BH,09H
PUSH BX

XCHNG CX, AX
________________________
________________________
________________________
________________________
________________________
RET
STACK_OPERATIONS ENDP
```

After ***PUSH AX***

Logical Address	Stack
________ : ________	
________ : ________	
________ : ________	
________ : ________	
________ : ________	
________ : ________	
________ : ________	
________ : ________	
________ : ________	
________ : ________	
________ : ________	
________ : ________	
________ : ________	

After Instruction = ________

AX = ________

BX = ________

CX = ________

SP = ________

After ***PUSH BX***

Logical Address	Stack
______ : ______	
______ : ______	
______ : ______	
______ : ______	
______ : ______	
______ : ______	
______ : ______	
______ : ______	
______ : ______	
______ : ______	
______ : ______	
______ : ______	
______ : ______	

After Instruction = ________
AX = ________
BX = ________
CX = ________
SP = ________

Question # 4: **(3+6 Marks)**

a. Write instructions to copy one STRING into another STRING in reverse order.

b. Write the following microinstructions:

1. **Read memory location addressed by AR and move the data to DR.**
2. **Clear the bit-3 in AL register using mask operations.**
3. **A register AL contains some value. Write instructions to find the 2's complement in AL.**

Question # 5: **(6 marks)**

Write down a program for the following:

Given: the last two digits of your SAP ID as hexadecimal digits e.g. 701223**12 has 12h.**
Write an assembly language program that consider the given value in BL then complement the bit-3 and bit-4 in BL. Use procedure BINOUT to display the binary value in BL.

Hint: Use mask operation to complement the selected bits.

Question # 6: **(8 Marks)**

Write an assembly language program where Data Segments contains a string consisting of words separated by blanks, ending with a "$".
Displays the string on a new line but with reversed word. You need to write a procedure that read the word from data section and reverse each word.
For example, "keep calm and keep solving" becomes "peek mlac dna peek gnivlos".

(Note: It can be implemented through stack or string operations. You can use any technique as your convenience)

Good Luck ☺

Nothing beyond this line will be evaluated